Effect of Quantization on Reliability of LLMs as Hallucination Detector in Medical Q&A

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April 09, 2025

B.A.Sc. Thesis



Abstract

Large language models (LLMs) offer powerful capabilities in medical question answering [1], but their high computational cost and propensity to hallucinate clinically inaccurate information pose serious risks in deployment. This thesis investigates how aggressive model compression—specifically k-bit quantization [2, 3]—affects the ability of LLMs to detect hallucinated versus truthful answers in a zero-shot binary classification setup. Using the MedHallu benchmark [4], we evaluate four instruction-tuned LLM families (Qwen2.5 [5], BioMistral [6, 7], Gemma [8], and LLaMA3-Med [9, 10]) across six GGUF quantization levels (q2, q3, q4, q5, q6, q8). Furthermore, we assess tradeoffs in performance (accuracy, precision, recall, F1, and abstention rate) and system metrics (inference latency, memory usage, and GPU load). Our findings reveal that quantization below 4–5 bits significantly degrades hallucination detection accuracy, especially for smaller models, while offering substantial gains in memory and speed.
Lastly, we also identify Pareto-optimal configurations that balance reliability with efficiency, providing actionable guidance for deploying compressed medical LLMs in resource-constrained clinical settings.
The code for this project is available at: https://github.com/nabilLearns/quantized-hallucination

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Chapter 1

Introduction

Large Language Models (LLMs) are achieving state-of-the-art performance across a range of natural language processing (NLP) tasks, including medical question answering [11]. However, these high-performing models often require vast computational resources—30 to 70 billion parameters and 128–280 GB of memory [11]—limiting their deployment in resource-constrained or privacy-sensitive environments such as rural hospitals or edge devices. To address these barriers, model compression techniques like quantization (reducing numerical precision) and pruning (removing parameters deemed less important) have been developed to enable smaller, faster, and more efficient models [12, 13]. While most prior work on compression has focused on preserving performance on general NLP tasks [12, 14], less attention has been given to the safety implications of compression—especially in high-stakes domains like medicine.

A key safety concern with LLMs is their tendency to generate hallucinations—statements that seem fluent and confident but are factually incorrect [15, 16]. In clinical settings, such errors can mislead users and compromise patient safety. Recently, researchers have explored a strategy that could be used to prevent or filter LLM hallucinations: instead of generating answers, LLMs can be repurposed as classifiers to detect hallucinations in model-generated text [17, 18, 19, 20]. This reframing treats hallucination detection as a binary classification task—given a medical question, a ground-truth answer, and a model-generated response, can another LLM accurately flag whether the response is hallucinated?

While these hallucination-detecting LLMs have shown promise, their utility in real-world workflows remains limited by size and latency. Quantization offers a potential solution by reducing

memory usage and accelerating inference [21, 22, 23]. Yet, the impact of quantization on hallucination detection performance—particularly in the medical domain—remains underexplored.
This thesis aims to fill that gap. We investigate how k-bit quantization affects the ability of instruction-tuned LLMs to detect hallucinations in medical question answering. We evaluate four LLM families—Qwen2.5 [5], BioMistral [7, 6], Gemma [8], and LLaMA3-Med [9, 10]—across six k-quantization levels (q2 to q8), using the MedHallu [4] benchmark dataset. The models are evaluated in a zero-shot classification setting, without fine-tuning, to assess their native capacity for hallucination detection under compression.
Our contributions are threefold:

- We quantify how k-bit quantization affects hallucination detection accuracy, recall, and abstention rates across hallucination types;
- We analyze system-level tradeoffs in latency, memory usage, and GPU load introduced by quantization;
- We identify Pareto-optimal tradeoffs between accuracy and efficiency, offering deployment guidance for medical LLMs in constrained environments.

By examining both classification performance and system efficiency, this work provides actionable insights for the safe and practical deployment of compressed medical LLMs.

Terminology and Notation

TP True Positives: Number of hallucinations correctly identified by the examiner model.

FP False Positives: Number of non-hallucinations incorrectly labeled as hallucinations.

TN True Negatives: Number of non-hallucinations correctly labeled as such.

FN False Negatives: Number of hallucinations incorrectly labeled as non-hallucinations.

Chapter 2

Literature Review

This chapter reviews existing literature that is relevant to **(1)** compression techniques for Large Language Models (LLMs), **(2)** the challenge of LLM-generated hallucinations, and **(3)** the emerging use of LLMs as hallucination detectors. The aim is to contextualize the research done in this thesis and clearly define the existing gaps this work seeks to address.

2.1 Efficient Processing of Large Language Models

Firstly, the successful deployment of LLMs in local-use settings (e.g. edge devices) depends largely on model efficiency and reliability.
Modern Large Language Models (LLMs) - often comprising tens to hundreds of billions of parameters – pose significant limitations for real-world applications, particularly in resource-constrained or privacy sensitive healthcare settings [11, 24]. LLMs can also pose environmental risks. The training and deployment of LLMs can result in the consumption of substantial natural resources (e.g., energy, water), as well as the emission of thousands of tonnes of greenhouse gases [13].
Various techniques have been developed to enable more efficient LLM deployments, including **(1)** model pruning, **(2)** knowledge distillation, **(3)** low-rank approximation, **(4)** neural architecture search, and **(5)** quantization [25, 26, 12].

2.1.1 Pruning Methods

Overview: Pruning methods reduce the number of parameters in a neural network. This is done by identifying and removing the LLM components (e.g., individual neurons, or larger components like attention heads or neural network layers) that least contribute to the models' performance. In doing so, pruning techniques can significantly reduce compute requirements whilst maintaining performance on a general benchmark [12].

Pruning methods can be distinguished by (1) their choice of component to prune (i.e., pruning granularity), and (2) their metric of choice (i.e., saliency metric) for determining which components are least important (and thus, the best candidates for pruning) [26].

Pruning Granularity: The 7 types of LLM components that a pruning method in literature can choose to focus on removing (either individually, or in a combination) are as follows: individual weights, activations, attention heads, embedding dimensions, sublayers, layers, or tokens. Examples include [27, 28] which removes weights only, [29] which removes both activations and attention heads, CoFi [30] which removes embedding dimensions, activations, attention heads and sublayers, [31] which removes layers, and LTP [32] which removes tokens.

Saliency Metric: To determine the LLM components to remove, pruning methods rely on a saliency metric that measures the importance of a component. For example, the magnitude pruning algorithm[27, 33] takes the magnitude of a weight $|\mathbf{W}_{ij}|$ as its measure of importance and removes weights with the least magnitude. A more modern approach, introduced by Sun et. al, [28] extends magnitude pruning by also taking the input feature activations $\mathbf{X}$ into account. Their approach, dubbed **W**eights **and a**ctivations (WANDA), measures the importance of each weight $\mathbf{W}_{ij}$ in a linear layer as $|\mathbf{W}_{ij}| * ||\mathbf{X}_j||_2$ [28].

2.1.2 Knowledge Distillation

Knowledge distillation (KD) involves training a smaller model (student) to replicate the behavior of a larger model (teacher). In the case of typical (non-LLM) deep neural networks, this is done by adding a loss term in the student models' training objective that measures the distance between student and teacher output logits (units in the final layer of a network with scores for each class ; in a computer vision application like object detection there could be different scores

for different potential objects, in LLMs, there is a score for each token in a vocabulary of tokens) [34, 26, 25]. In the context of LLMs, the student can try to learn from other aspects of the teacher as well, including its hidden states, attention maps, and token embeddings [26].
If successful, KD enables the usage of the student model for faster inference, and lower memory requirements, with essentially the same performance characteristics as the more complicated teacher model [34, 25]. This technique can be combined with other compression techniques, such as pruning or quantization [34, 35]. Still, KD methods are a relatively unpopular compression technique, as successful application requires long training times [26]. A long training time requirement translates to high resource cost, so it is unlikely that this compression technique would be prioritized for the deployment of LLM(s) in resource-constrained environments.
Lastly, although effective in reducing model computation requirements, the effect of knowledge distillation on safety and hallucination risks remains relatively unexplored [16].

2.1.3 Low-Rank Factorization

Low-rank factorization techniques replace large weight matrices into products of smaller matrices. This generally results in lower memory requirements, for little sacrifice in modal accuracy [25]. However, its impact on medical question answering and hallucination risk has yet to be extensively studied [16].

2.1.4 Neural Architecture Search (NAS)

Neural architecture search automates the discovery of optimal network architectures that achieve high performance with lower computational overhead [25]. NAS techniques have potential benefits for efficient deployment but require significant computational resources for training and searching architectures, posing limitations for stakeholders with limited resources.

2.1.5 Quantization

Quantization is a popular compression technique that involves representing model parameters and activations with reduced numerical precision (e.g., 8-bit integers instead of 32-bit floating points), leading to significant memory and computational savings [21, 36]. Quantization has become a

standard industry practice, especially 8-bit linear quantization [37]. Recent advancements and popular frameworks include k-quantization for CPU-centric inference [3, 2], BitsandBytes [21], Activation-aware Weight Quantization (AWQ) [22], and GPTQ [23]. Despite these advances, the implications of quantization on LLM safety, particularly in relation to hallucination detection, are, to the best of our knowledge, relatively unexplored [16]. This is a gap this thesis seeks to investigate.

2.1.6 Perplexity

Generally, papers that compare compressed LLMs use general measures of model quality, like accuracy or **perplexity** as their criterion of comparison [26, 12]. Now, whilst accuracy is a metric applicable to most machine learning models (e.g., classification, NLP, computer vision models), the use of perplexity is unique to language models. At a high level, perplexity measures how good a language model is at predicting the next n tokens of a given text [38]. This NLP task is known as ”next sentence prediction” [39]. Many NLP tasks, like sentence translation, summarization, and question answering can be formulated as ”next sentence prediction” problems [38, 39]. The formal definition of perplexity and its explanation [38] are below:

$$\text{Perplexity} = \sqrt[n]{\prod_{i=1}^{n} \frac{1}{P(t|t_0, ...t_{i-1})}}$$

For a given text with $T = g + n$ tokens, with g tokens given to the model as context (i.e. in a prompt), and the next n tokens representing the 'correct' tokens that the model is tasked with predicting, the likelihood of the model correctly predicting the next n tokens (assuming they are i.i.d.) can be modeled as $P(t_0, ..., t_n) = P(t_0) * P(t_1|t_0) * P(t_2|t_0, t_1) * ... * P(t_n|t_0, ...t_{n-1}) = \prod_{t=1}^{n} P(t|t_0, ..., t_{n-1})$. So, if the probability $P(t_0, ..., t_n)$ of correctly predicting the next n tokens is high, then the overall perplexity value is low. A lower perplexity value is better.

Explanation of perplexity via example To further illustrate the meaning of the equation via an example, say an LLM is only given the first two tokens of the following sentence:

> The quick brown fox jumped over the lazy dog,

The Gemma3 tokenizer would break this sentence down into 10 tokens, as shown in the table

below [40]:

Input	The quick brown fox jumped over the lazy dog									
Tokens	`<bos>`	`The`	`quick`	`brown`	`fox`	`jumped`	`over`	`the`	`lazy`	`dog`
Embeddings	2	651	4320	8426	25341	32292	1163	573	27894	5929

Table 2.1: Token embeddings example

In the context of this thesis, perplexity, though widely adopted as a comparison criterion for compressed LLMs, does not directly inform users about whether a particular LLM would be safe to use (i.e. with respect to its risk of hallucination). LLM safety, as it relates to this thesis, is discussed in further detail below.

2.2 Safety Risks in LLMs

Despite their advancing capabilities, the successful deployment of LLMs in sensitive settings like medicine or finance is impeded by safety and reliability issues [13, 41]. Of the numerous safety risks associated with the deployment of LLMs, hallucination is a major one and a central focus of this thesis.

2.2.1 Hallucination

Hallucinations - statements produced by LLMs that seem plausible but are factually incorrect [15, 38] - are a critical safety concern in deploying LLMs in sensitive (e.g., medical, finance) applications [16].
Not all hallucinations are the same, and researchers have proposed various ways to categorize hallucinations into different types. Weng [15] categorizes hallucinations into one of two types:

- **Intrinsic Hallucination:** When the LLM output contradicts factual information within the context provided to it (e.g., input prompt: "").
- **Extrinsic Hallucination:**

On the other hand, the MedHallu benchmark [4], which forms a major component of this thesis, defines four categories of hallucination. These are adapted from the general categories used by

the KnowHalu benchmark [42] to categories that fit better within a medical setting. The four hallucination categories are as follows:

•

We draw on this categorization in our analysis of our experimental results.
Mechanism Misattribution:
Incomplete Information:
Question Misinterpretation:
Evidence Fabrication:
[15] Defines two categories of hallucination: (1) Instrinsic, and (2) Extrinsic hallucination. In the former case, hallucination occurs when there is inconsistencies between the context provided in the input prompt and the output produced by the LLM. Whereas the latter form of hallucination occurs when the LLM output is not consistent with 'external world knowledge', that it would be expected to know from its pretraining data (e.g., if an LLM trained on data from 2021 was asked to name the current prime minister of Canada, any response other than something along the lines of "I do not know" would be an extrinsic hallucination).

2.3 Side effects of Compression

Works on LLM compression can qualify their techniques by showing that 'side effects' of compression are low. Existing work typically focuses on holistic measures of performance, such as maximizing compression * reduce inference time, model size * maxmimize compression ratio, speedup ratio [12]
minimizing side effects * reasoning ability * Eleuther AI LM harness
Existing work on LLM compression methods typically qualify their technique by comparing the overall accuracy of compressed models with their original uncompressed counterpart.
However, there is not much that focuses on safety-related side effects of compressing LLMs. Here are some works that exist:

2.3.1 toxicity, hallucination risk, bias, etc.

- most focus on accuracy, not much focus on safety/reliability
- some works explore effect of pruning ; but not quantization ;
- NLP tasks that hallucination detection has been applied to ; text-summarization; this is easy to check for instrinsic hallucination(?) ; not much done in Q&A, let alone medical Q&A?

Our study focuses on the side effect of hallucination, in LLMs compressed via quantization, in the NLP task of question answering, on a medical dataset.

2.4 LLMs for hallucination detection

Recent works like [17, 18, 43, 19] have used LLMs to detect hallucinations in model-generated output. Cohen et al. [18] propose an interactive "LLM-examiner" approach in a question-answering task where an LLM-examiner interactively engages with an LLM-answerer by asking it a question, asking follow-up questions based on the LLM-answerers' answers, and then making a decision about whether the LLM-answerer made a factual or hallucinated response to the originally asked question. Their work primarily uses models with around 7 billion (7B) parameters, and make use of both open-source and private models like GPT-4 [44] in their evaluations. We do not study private LLMs (e.g. Claude LLMs [45], GPT-4) because their model weights are not publicly accessible. Without model weights, we cannot apply compression techniques (e.g. quantization, the focus of this thesis) to study their performance profile at different compression levels. CH-Wang et al. [43] propose an LLM-examiner that inspects the internal components of an LLM-answerer to determine whether it hallucinated.

2.5 Research Gap and Motivation

Our review identifies a research gap in understanding how quantization influences the ability of an LLM-examiner to detection hallucinations in LLM-produced responses to medical questions.

This is an important gap to address, given the sensitive nature of healthcare applications, the potential consequences of deploying LLMs which produce unreliable (i.e. hallucinated) responses (e.g., to medical questions) and the dual objective of making local-deployment of LLMs practical in the interest of preserving private information. Our research explicitly addresses this gap, by analyzing model and system performance tradeoffs that arise when quantizing LLM-examiners of hallucination in medical QA. We hope that the results of this study can provide practical guidelines for medical applications. Our work addresses previously unexplored dimensions of the impact of quantization on LLM reliability in medical contexts.

Chapter 3

Methods

This chapter outlines the methods used to evaluate the effects of quantization on hallucination detection performance in large language models (LLMs) applied to medical question answering. We describe the task formulation, dataset, experimental setup, models and quantization schemes, prompt design, evaluation metrics, and limitations. Our goal is to provide sufficient detail for replication while explaining the rationale behind our choices.

3.1 Task Formulation

We frame hallucination detection as a zero-shot binary classification task. A language model is presented with a prompt containing a medical question, a ground-truth answer, relevant context (usually from a PubMed abstract), and a candidate answer that may be hallucinated. The model must respond with either `1` (hallucinated) or `0` (not hallucinated). This setup follows the format introduced in the MedHallu benchmark [4].

We focus on the medical domain due to the high risk associated with misinformation in healthcare applications. Unlike casual use cases, hallucinations in medical LLMs can mislead clinicians or patients. However, most existing evaluations of LLM hallucinations focus on generative metrics in general-purpose or summarization settings [15, 33]. Our work is novel in that it evaluates classification-based hallucination detection in compressed medical LLMs.

3.2 Dataset: MedHallu

Our experiments are conducted using the MedHallu dataset [4], which contains 10,000 examples constructed from the PubMedQA dataset [46]. Each example includes:

- A medical question,
- relevant background context (from a PubMed abstract),
- a ground-truth answer, and
- a hallucinated answer.

Hallucinations were generated using the Qwen2.5-14B model with structured prompting. All hallucinated answers are categorized by difficulty (easy, medium, hard) and one of four hallucination types: misinterpretation, incomplete information, mechanism/pathway misattribution, and methodological evidence fabrication.

3.2.1 Limitations in Labeling and Categorization

While the dataset offers fine-grained labeling, we note that some examples may be miscategorized. For instance, a hallucination labeled as a ”Misinterpretation of Question” may instead be better described as a ”Mechanism Misattribution,” based on its factual contradiction rather than its relevance. These discrepancies suggest the need for more precise annotation criteria and reinforce the importance of analyzing performance by category. Case studies illustrating this are discussed in the Discussion chapter.

3.3 Prompting and Labeling

We follow the prompting approach of [4]. Each prompt includes a clear instruction asking the model to assess the validity of the answer and respond with a binary classification. An example prompt [4] is shown below:

```
Question: Can surgeon familiarization with current evidence lead to a
change in practice?
```

```
Context: Despite evidence against its utility, many surgeons continue...
Ground Truth Answer: This study has shown that mailing out a summary
of current evidence to surgeons concerning a certain issue is not sufficient
to lead to a change in practice.
Candidate Answer: Mailing a summary of current evidence to surgeons
led to a statistically significant reduction in the use of prophylactic
nasogastric decompression in elective colonic resections.
Instruction: Determine whether the candidate answer is hallucinated or
correct. Reply only with a single number: 0 (not hallucinated) or 1
(hallucinated).
```

Model responses are parsed by examining the first 10 characters. If they contain a valid binary digit (0 or 1), the result is recorded. Otherwise, the prompt is marked as abstained. This parsing strategy ensures consistency and minimizes false interpretations of verbose outputs (e.g., from BioMistral).

3.4 Model Selection

We evaluate four families of instruction-tuned LLMs:

- **Gemma3 1B, 4B, 12B**;
- **Qwen2.5 0.5B, 1.5B, 3B, 7B**;
- **BioMistral 7B** (medically fine-tuned);
- **LLaMA3-Med 8B** (medically fine-tuned).

Instruction-tuned versions were prioritized over base models for better task adherence [47]. Model sizes were selected to explore a diverse range of parameter scales and to include both general-purpose and medically-specialized instruction-tuned LLMs. We chose open-source models available in GGUF format for compatibility with quantization-aware runtimes.

3.5 Quantization and Runtime Considerations

We analyze performance across six levels of GGUF quantization: q2, q3, q4, q5, q6, and q8. Each level reduces the number of bits used to represent model weights (e.g., q2 uses 2-bit weights, q8 uses 8-bit weights), thereby decreasing memory usage and improving inference speed. Quantization is performed using data-independent, weights-only k-bit quantization following [2]:

$$Q = \text{round}\left(\frac{X}{s}\right), \quad \text{where } s = \frac{\max(X)}{2^{k-1} - 1} \tag{3.1}$$

This format is compatible with GGUF, which supports sub-block partitioning and higher-bit overrides (e.g., q2_K_XL means a large portion is quantized at greater than 2-bit precision). GGUF is optimized for local inference and popular in consumer-grade LLM deployments.
Quantized models were run using the Ollama runtime with the Open WebUI template on Vast.AI GPU instances [48]. Hugging Face dequantizes weights on load, and 'llama-cpp-python' introduced excessive latency. We confirmed dequantization via system logs (Figure A.1 in Appendix), emphasizing the need for quantization-aware runtimes to preserve efficiency gains.
This choice aligns with system-algorithm co-design principles [36, 37], where software (runtime) must complement hardware and model compression to achieve low-latency, low-memory deployment.

3.6 Evaluation Metrics

We track both model-level and system-level metrics:

3.6.1 Model Performance

- Accuracy, Precision, Recall, F1-Score
- True Positive Rate (TPR), False Positive Rate (FPR)
- Abstention Rate (proportion of invalid or verbose LLM outputs)
- Breakdown of TPR and FPR by hallucination category and difficulty

3.6.2 System Performance

- Latency: min, max, average, standard deviation, total experiment time
- GPU metrics: GPU load (max, average, standard deviation), GPU memory utilization (max, average, standard deviation), total available memory
- Throughput (prompts/second)

GPU statistics were collected using the GPUtil library [49], and latency was recorded from the start of prompt submission to output parsing. Total experiment time and cost were estimated by parsing latency logs and multiplying by the runtime cost (CAD $0.698/hr). Full results and script references are included in the Appendix.

3.7 Hardware Setup

Initial development was performed on V100 GPUs for cost-effective debugging. This choice was made because inference with quantized models (e.g., bitsandbytes) required CUDA support and could not run on local CPU. Final experiments were run on NVIDIA A40 GPUs using the Vast.AI Open WebUI (Ollama) template. All experiments were run on a single GPU instance, and no parallelism was used during inference. The runtime cost was approximately CAD $0.698/hr, as shown in Appendix A.4.

3.8 Baselines

3.9 Limitations

- Limited to one dataset (MedHallu), scoped to medical QA tasks
- Results may not generalize to other NLP tasks like summarization or domains like legal or financial QA
- Experiments run only on GPUs; CPU and edge-device (e.g., on an NVIDIA Jetson Nano) performance was not profiled (but can be for future work)

- Only weights were quantized; quantization methods (e.g., GPTQ [23], AWQ [22], LLM.int8() [21]) that include activations in their quantization scheme (e.g., W8A8 [36]) were not evaluated.
- Other quantization methods (e.g., GPTQ, AWQ) were excluded for compatibility reasons
- GGUF quantization's practical efficiency depends heavily on runtime, as shown by dequantization logs

3.10 Summary

This methods section documented the experimental design and decisions made to assess hallucination detection in quantized medical LLMs. Key contributions include a multi-model, multi-bit benchmark using GGUF quantization, and analysis of tradeoffs between compression and classification reliability. The full experiment setup, data, and code are available at the following link for reproducibility: https://github.com/nabilLearns/quantized-hallucination. Details on parsing scripts and cost estimates are in Appendix A.6, and A.5 respectively.

Chapter 4

Results

4.1 Overview

This chapter presents the experimental results evaluating the impact of K-bit quantization (also referred to as K-quantization) on various Large Language Models (LLMs). We first examine the direct effect of varying K on standard model performance metrics (e.g., accuracy, precision, recall), and system-level performance metrics (e.g., peak memory usage, latency). Afterwards, we examine the tradeoffs between model and system performance, and use Pareto dominance analysis to identify optimal model configurations.

4.2 Impact of K-Quantization on Performance

4.2.1 Model Performance Metrics

Accuracy measures the proportion of judgements made by the LLM hallucination examiner that were correct [39, 50]. Because our dataset is balanced (i.e., has an equal proportion of 1000 ground truths, and 1000 hallucinated answers to classify), accuracy can be used to get a rough understanding of overall classifier quality [51].

Figure 4.1 shows how classification accuracy is affected as quantization level varies. For ease of comparison, we place the subplots for general-reasoning models (e.g., Qwen2.5, Gemma3) side-by-side, and do the same for Llama3 and Biomistral, the model families fine-tuned on medical

datasets.

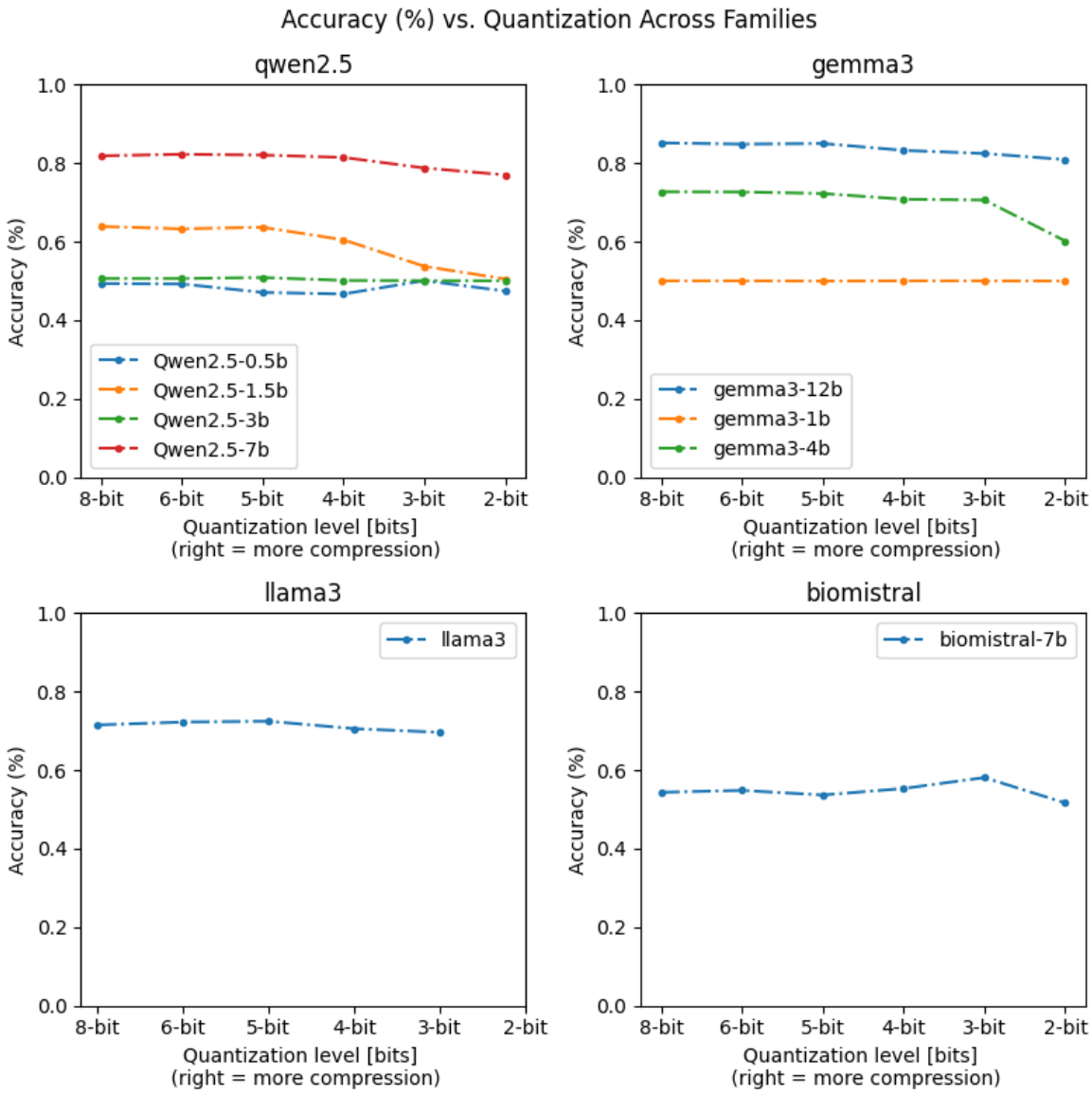


Figure 4.1: Model accuracy as a function of K-quantization level. Lower K values (towards the right) mean that fewer bits are being used to represent each weight $\mathbf{W}_{ij}$ in a set S of quantized weights within the LLM.

Reducing precision can negatively affect accuracy. The plots above show how the side effects (in this case, reduced classification accuracy) of reduced precision varies across different model families.

3 smaller models often perform near or below random chance The upper two plots show the

worst performing models (with respect to accuracy) amongst the tested model families: Qwen2.5 0.5B, 3B, and Gemma3 1B. Quantization has little effect on the classification accuracy of these small ($\leq$ 3B parameter) language models. In particular, the Qwen2.5 3B and Gemma3 1B models exhibit a static accuracy of around 50% over the six tested quantization levels. the Qwen2.5 0.5B parameter model shows similar behavior, except that it shows worse-than-baseline accuracy levels at the 2,4, and 5-bit quantization levels. Because the MedHallu dataset comprises an equivalent number of ground truth and hallucination answers (

$$n = 1000$$

), the probability of randomly drawing either a ground truth or hallucinated answer from the dataset is equivalent. Thus, the 50% accuracy of the small Qwen2.5 and Gemma3 models is the same that can be expected from a ”random guess” baseline model. A possible explanation for this naive guessing behavior is that the capacity of these smaller models is too small in comparison to the complexity needed to handle the task of understanding the medical question, context, candidate answer, and instruction provided within the prompt. If the standard prompt was too difficult for the smaller LLMs to comprehend, further work could explore techniques to simplifying the prompt given to relatively small LLM-examiners. Drawing upon a recent study conducted by Jain et al. [52], an LLM like GPT-4 could measure the difficulty of a prompt and validate whether a simplified prompt is less difficult than the original. Further experiments could be run to see if relatively small LLMs perform better with prompts that are easier to comprehend. This however, does not seem scalable for a real-world use case where an LLM may be used for on-the-fly question answering, and there is little opportunity to experiment with prompt simplification.

3 mid-sized models showed performance degradation with increased quantization Qwen2.5 1.5B, Gemma3 4B, and the Biomistral 7B models show better than naive baseline accuracy levels, but a decline in performance when quantized to a greater extent. (1) In particular, Qwen2.5 1.5B performd a little above 60% accuracy from 8-bit to 5-bit quantization levels, but gradually declinrd to around naive guess performance levels at the 2-bit level. (2) The BioMistral model showd similar behaviour, with performance a little under 60% accuracy from 8-bit to 4-bit baselines, followed by instability at the 3-bit level, and a degradation in performance at 2-bits. (3)

Lastly, the Gemma3 4B model hovers around 0.75% accuracy, with a sharp decline to around 60% at the 2-bit level.

3 larger models show more resilience to quantization The Llama3, Qwen2.5 7B, and Gemma3 12B parameter models showed a slow decreasing trend in accuracy as the number of bits used to represent quantized weights decreases.

Observations from comparing models Generally, the highest performing models were the general-reasoning models (Qwen2.5, Gemma3) with ≥ 7B parameters. This may have been possible because we include relevant medical knowledge to the question asked in the prompt given to the classifier, which may remove the advantage that medically-fine tuned LLMs like Med-Llama and Biomistral have over the general reasoning models. In a different prompt setting (e.g., where no context is given in the prompt), it is possible that the medically-fine tuned models may perform better. This can be verified with future experiments. Lastly, it is noteworthy that the Qwen2.5 1.5B model consistently outperforms its 3B parameter counterpart, which behaves like a random-guess baseline model. A limitation in our comparison of models is that for the medically fine-tuned LLMs, we only had access to one model size. Unlike the general reasoning models, this limits the depth of insight we can gain into the performance degradation profile for a medically fine-tuned model family.

Still, due to the limited scope of the experimental environment (i.e., perhaps there can be different behaviour if a different prompting strategy is used ; we included contextual knowledge relevant to answering the medical question within the prompt for all LLMs ; if no context was provided perhaps the results would be different; Still, this would be more of a test of what's contained within pretrained knowledge, than their general reading comprehension ability; in that case, we could learn about, for an LLM which is assumed to have lots of medical knowledge ingrained in it (i.e., from pre-training), what extent does quantization remove that knowledge from the LLM; this is a future direction of work, that can be explored to reveal further insights into the behaviour of performance degradation in LLM-examiners), it is not known if this observation holds for LLM families that were not included in this experiment (e.g., private models like Claude 3.7 Sonnet [45] have already been used to help clinicians make diagnoses [53], and some have passed USMLE-level examinations [1]; these models would be excellent candidates for testing the effects of quantization due to their relevance to physicians; however such a study only

seems to those who have access to the weights of those models e.g., certain employees). Yang et al [53] studied the use of closed-sourced LLMs like Gemini [54, 55] and GPT-4 [44] for coming up with medical diagnoses. They did four repeated prompts to an LLM, and retried up to two times if an LLM did not respond [53]. They categorized LLM-produced hallucinations into 5 categories, which differ from those used by [4] in MedHallu. They studied LLMs in the context of producing medical diagnoses to challenging cases, whereas MedHallu [4] primarily comprise questions related to scientific medical publications.

Precision, defined as $(TP)/(TP + FP)$ (see introduction, section 1), measures the proportion of correct positive predictions – instances the LLM correctly identified as hallucination – out of all instances the model *predicted* were hallucinations [39, 50, 51]. A high precision value indicates that when the model flags an answer as a hallucination, it is likely correct, signifying fewer false positives.

Figure 4.2 illustrates how classifier precision changes across different k-quantization levels for the selected LLM families.

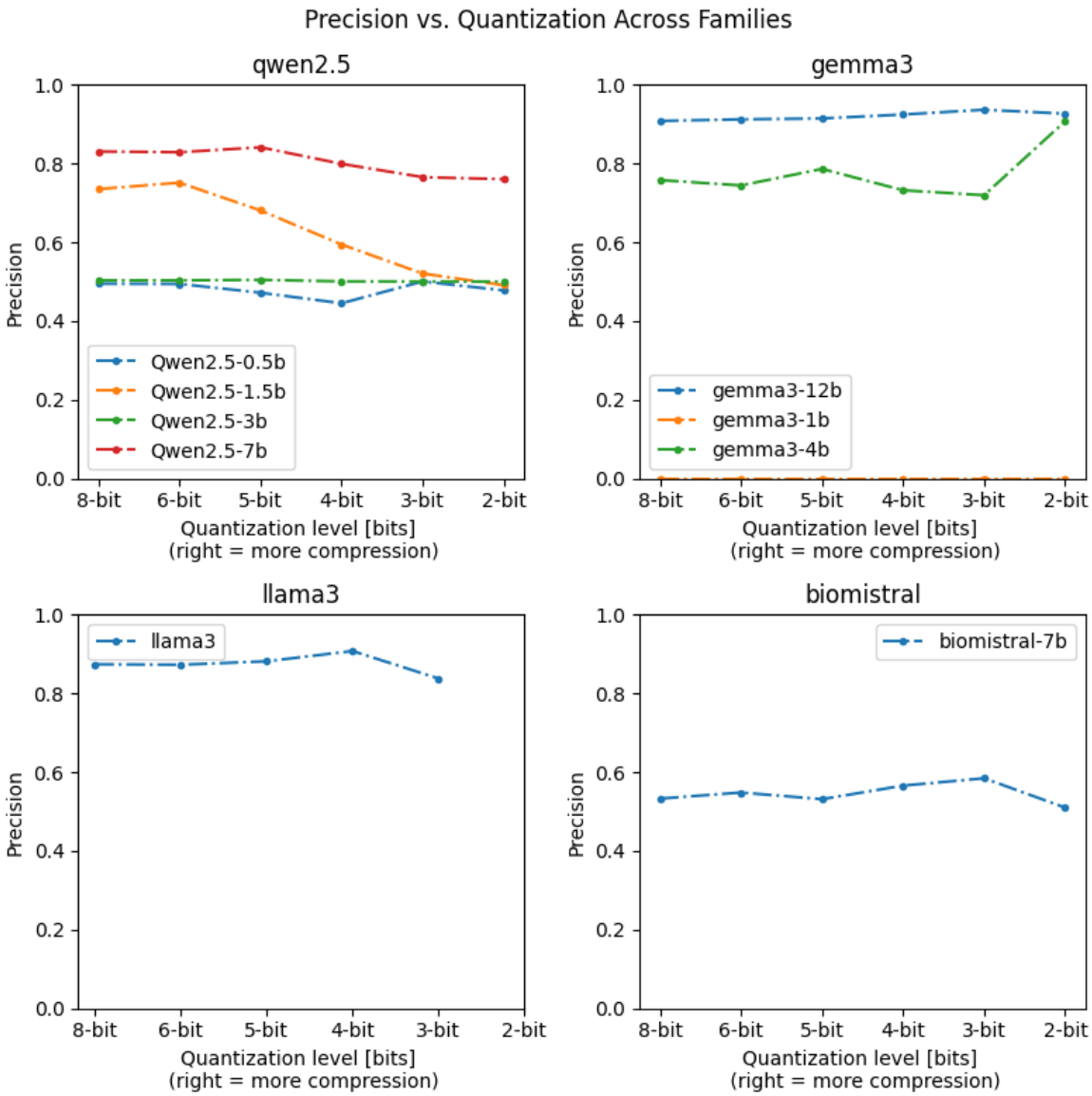


Figure 4.2: Precision across K-quantization levels for two general-reasoning model families (qwen2.5 and gemma3), and LLMs that were fine-tuned on medical data (Llama3 [9, 10] and Biomistral [7, 6]).

Evaluating the precision trends in Figure 4.1, we find that precision performance under k-quantization varies significantly based on the specific LLM family and size.

Biomistral-7B: The precision trend for Biomistral closely mirrors its accuracy pattern. We observe an slow increase in precision from 8-bit to 3-bit quantization (precision rising from approximately 0.55 to 0.60), followed by a sharp decline from 3-bit to 2-bit quantization levels. This

parallel behavior suggests that Biomistral's overall corectness at different quantization levels is strongly linked to its tendency to generate false positive predictions.

Llama3: Consistent with with its stable accuracy performance, the medically fine-tuned Llama3 model shows robust precision across the tested quantization range, generally staying above 0.8. It reaches a peak precision near 0.9 at around k=4, before showing degradation at from the 4-bit to 3-bit quantization levels. This indicates that Llama3's positive classifications (identifying hallucinations) are reliable and relatively unaffected by GGUF quantization down to 3 bits. Note, that due to time and resource constraints, as well as technical difficulties with the Llama3-med Q2_k model (inference seemed to hang indefinitely after a seemingly arbitrary number of examples) we were not able to complete an experimental run for the 2-bit quantization level.

Qwen2.5 Family: Precision degradation within the Qwen2.5 model family clearly depends on model size:

- The smallest model (0.5B) achieves precision near 0.5, comparible to its accuracy, suggesting performance close to random chance with respect to false positives
- On the other hand, the 1.5B model shows a marked *decrease* in precision as K decreases, descending from roughly 0.75 at k=6 to 0.48 at k=2. This trend is more pronounced than its accuracy degradation, suggesting that quantization levels with fewer bits have significantly increased tendency to incorrectly flag ground truth answers as hallucinated.
- The 3B model, which maintains a precision near the 0.5 baseline for all K values, exhibits a performance degradation profile similar to its accuracy profile shown in Figure 4.1.
- The largest (7B parameter) Qwen2.5 model maintains high precision (between 0.83 to 0.78) with a slight downward trend as K decreases. Its precision generally exceeds its accuracy across most K values, suggesting that while it may fail to identify some actual hallucinations (impacting recall), the instances it *does* flag as hallucinations are usually correct.

Gemma3 Family: Like the Qwen2.5 family, the Gemma3 models show size-dependent performance degradation profiles:

- Notably, the Gemma3-1B model has a precision of 0 across all quantization levels. Although its accuracy hovered around 50% (random chance), this near-zero precision reveals

that either (1) none of its positive predictions were correct or (2) it did not make any positive predictions in the first place, and instead classified all instances as ground-truth. Further investigation of the outputs produced by the 1B model reveal that the latter case was true.

- The 4B model has precision around 0.78 from the 8-bit to 3-bit quantization levels, but exhibits a substantial increase in precision from k=3 (around 0.75) to k=2 (approximately 0.9).

- The 12B model beings with exceptionally high precision at k=8 (about 0.95), and remains at a similar performance at lower quantization levels.

Summary of precision findings: In summary, large models such as Llama3-8B, Qwen2.5-7B, and Gemma3-12B generally show higher and more stable precision, though the Gemma3-4B model shows some instability at bit levels around k=3 and k=5. On the opposite end of the performance spectrum, the plot for Gemma3 1B highlights a failure mode for the model: it totally avoids making positive predictions (assumes all answers are ground truth). Qwen2.5 1.5B model shows clear degradation in precision at lower bit levels, whereas Gemma3 4B shows a suprising sharp increase in precision at the lowest bit-width.

Recall, defined as $(TP)/(TP + FN)$ (see introduction, section 1), measures the proportion of hallucinated answers that the LLM-examiner was able to detect. Having high recall is important as it signifies that the LLM-examiner misses few hallucinations.

Figure 4.3 shows how classification recall is affected by the extent of k-quantization.

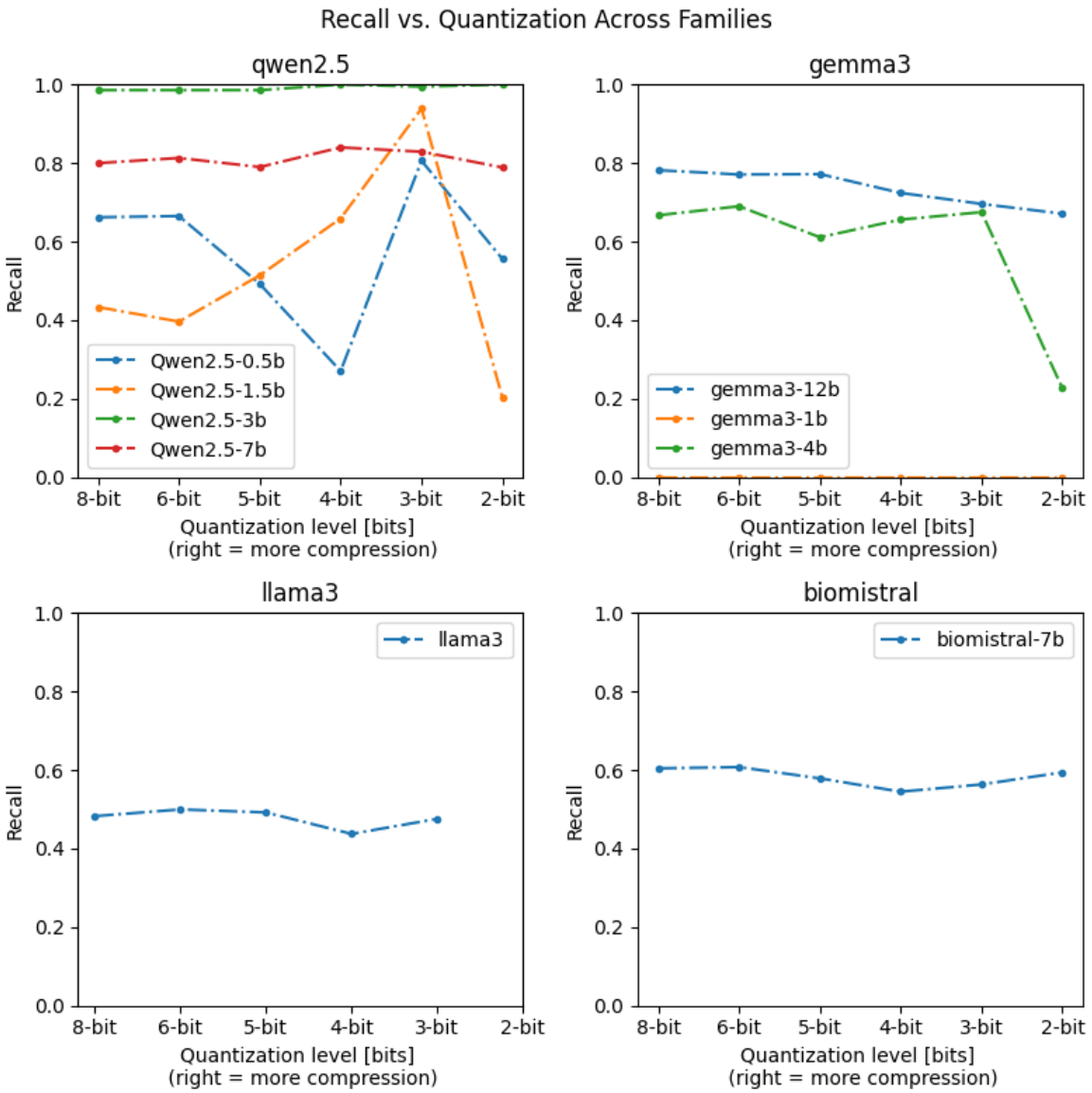


Figure 4.3: Recall across K-quantization levels.

Recall performance profiles are diverse across the 4 tests model families: **BioMistral-7B:** The recall curve for BioMistral-7B indicates a moderate ability to detect hallucinations (around 60% were detected of the 1000 hallucinated answers). Its recall is relatively stable across the different quantization levels, with peak performance at 8-bit and 2-bit levels.

Llama3 8B: This exhibits a similar recall-quantization profile shape to its medically fine-tuned counterpart (BioMistral-7B), with recall slowly declining from the 8-bit to 4-bit levels, followed by an increase for lower bit-widths. However, unlike BioMistral-7B, this models' recall is more

prone to missing actual hallucinations.

Qwen2.5 Family:

- The 0.5B model is best at recognizing hallucinations when 3-bit k-quantization is applied. It has a highly erratic recall profile, with a recall of 0.64 at the 8-bit and 6-bit levels, followed by a decline to 0.3 at the 4-bit level, and a spike in performance for the lowest two quantization levels.

- The 1.5B model shows a clear increasing trend in recall performance as more extensive quantization is applied, increasing from 0.43 at the 8-bit level to 0.93 at the 3-bit level. This model does not bode well with 2-bit quantization, as its recall degrades to 0.2.

- The 3B model gradually increases from 0.98 recall at the 8-bit level to a perfect recall at the 2-bit level. In light of its stable 50% accuracy from Figure 4.1, this indicates that the model tends to assume nearly, if not all answers it is given is a hallucination. With a stable accuracy, precision, and recall over the different quantization levels, it seems that k-quantization has little effect on the 3B models performance. Still, its naive guessing strategy makes it unreliable for use in a real-world setting.

- With a recall hovering around 0.8 at all quantization levels, the 7B model shows relatively stable performance. With its best recall at the 4-bit and 3-bit levels (of 0.84 and 0.82 respectively), the 7B model performs better than its smaller counterparts, even when they are compared at higher bit-levels. Performance from 8-bit to 2-bits: 0.8, 0.81, 0.79, 0.84, 0.82, 0.78

Gemma3 Family:

- With 0 recall, 0 precision, and 50% accuracy across all quantization levels, the 1B parameter Gemma3 model is not able to detect hallucinated answers. Instead, it assumes that every answer it is presented is ground-truth. Given its relatively small size, it is possible that the model may be too small to learn or tackle the classification task, even at full bit precision.

- The 4B variant of the Gemma3 series is considerably better at detecting hallucinated answers, with a detection rate of around 60% from the 8-bit to 3-bit quantization levels. The

3-bit quantized variant would be the best option for deployment, given that it performs better than the other quantization levels and has been quantized to the greatest extend (with reasonable performance).

- The 12B parameter Gemma3 model has the highest recall overall, maintaining 0.78 - 0.75 recall across all quantization levels. The shallow slope that the plotted points follow indicates a strong resilience to quantization, and that the model likely retains sufficient capacity even after compression.

In general, model size plays a critical role in its robustness to quantization. Smaller models (e.g. $\leq$ 3B parameters) show highly erratic behavior, whereas larger models ($\geq$ 7B parameters) show relatively stable performance across all quantization levels (as indicated by their relatively shallow slope). Large general reasoning models show the best performance amongst the 4 tested model families, with a slow decline with increasing extent of quantization. The medically fine-tuned models on the other hand, have a pattern of slow decline in performance from 8-bit to 4-bit levels, followed by a surprising increasing trend in recall at lower bit quantization levels.
The F1-score, plotted in Figure 4.4 is the harmonic mean of precision and recall, providing a single metric that balances both aspects. While often used for imbalanced datasets, it serves here as an additional measure. The F1 trends generally follow accuracy (see Figure 4.1) confirming that for this task with a balanced dataset, accuracy can be a primary indicator of model quality. The models with highest F1-score align closely with those showing the highest accuracy at each k-bit quantization level.

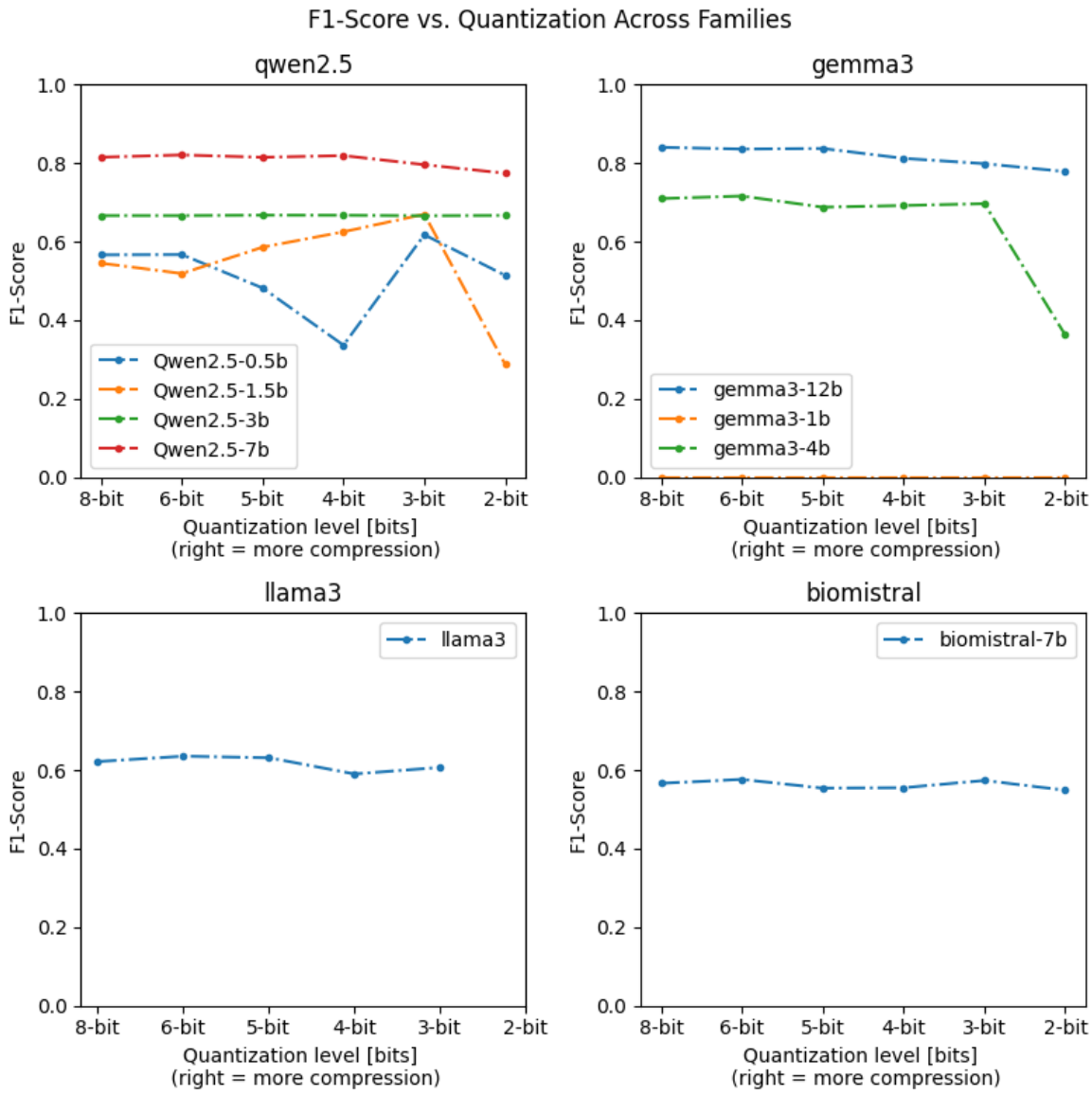


Figure 4.4: F1-Score across k-quantization levels.

The F1 trends shown in Figure 4.4 above mirror the accuracy trends in Figure 4.1. In both cases, the largest general-reasoning models (e.g., Qwen2.5 7B, Gemma3 12B) are the top performers, and maintain relatively stable F1-scores across the different quantization levels.

Like the large general-reasoning models, the medically fine-tuned Llama3 and BioMistral models show the most robustness to quantization with moderate yet consistent scores (around 0.6) across all bit-levels.

Mid-sized models such as Gemma3 4B show degradation in the F1-score after the 3-bit level,

whilst smaller models (e.g., Qwen 1.5B, 0.5B) show highly erratic performance.
These patterns reinforce the insight that model size and architecture influence resilience to quantization.

4.2.2 Abstention Behaviour

Abstention provides insight into the ability of the model to comply with output formatting instructions in a given prompt. A high abstention rate can indicate a tendency to produce invalid (e.g. overly verbose) outputs. This can be caused by an inability to follow prompt instructions. It can also mean that the LLM is frequently unsure of whether to classify a given medical answer as either hallucination or ground truth.
Figure 4.5 examines the abstention rate of the four LLMs used in our study. The abstention rate answers the following question: of the 2000 answers each model was asked to classify, what proportion did the model produce a valid output?

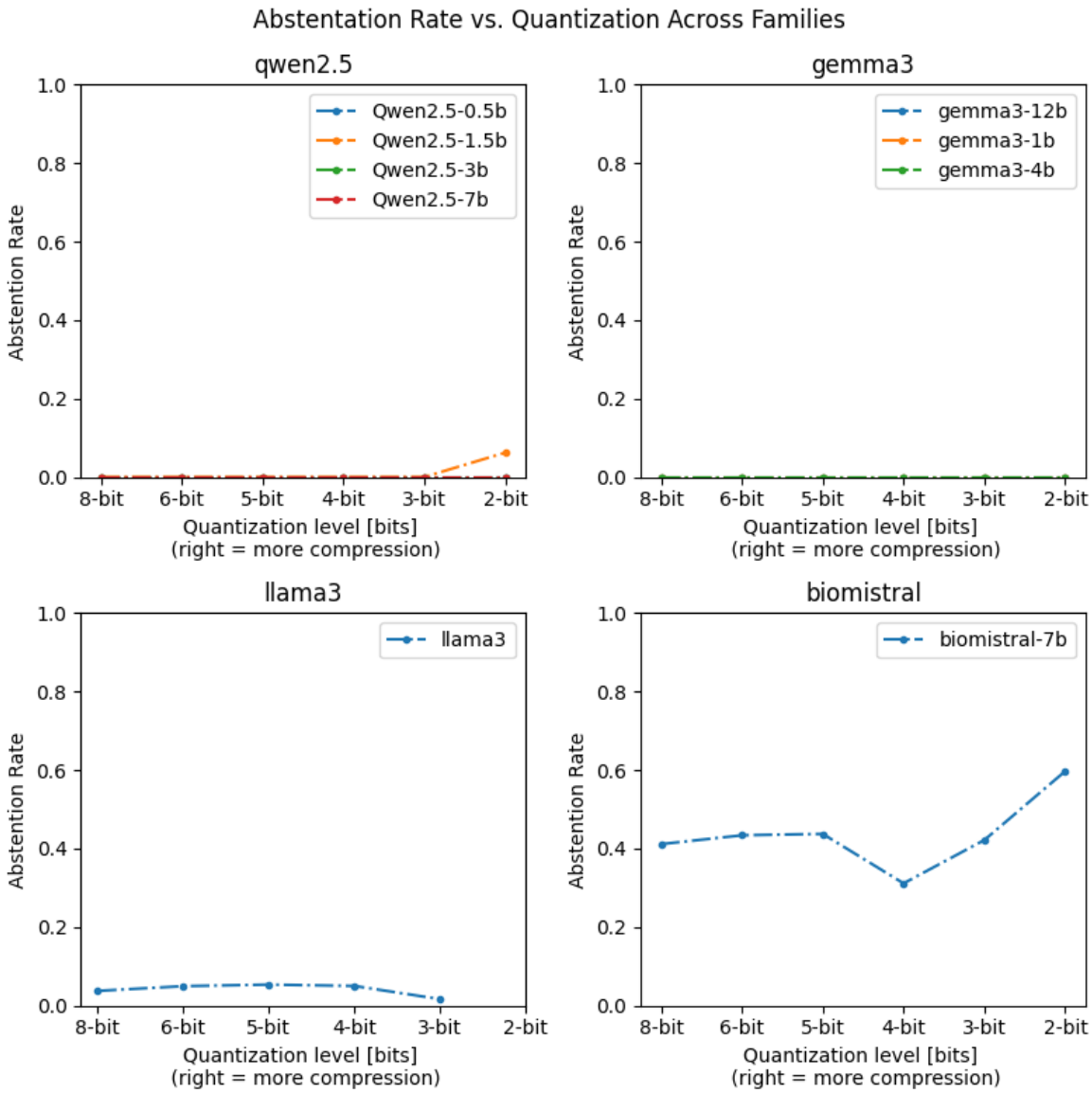


Figure 4.5: Abstention rate across different K-quantization levels.

The general-reasoning models have the lowest abstention rates (of 0) and show consistent performance across all quantization levels. An exception is the Qwen2.5 1.5B model, which shows an increase in abatention rate at the 2-bit level. These models are instruction-tuned models, which may explain their ability to consistently produce valid outputs [47].

As for the medically fine-tuned models, they generally have higher abstention rates (this makes sense, as they are not instruction-tuned). The reason for abstention can be explained by inspecting their output logs, which highlight that these models frequently produce verbose output (e.g.

paragraph-long explanations for their reasoning, even though the prompt only asks for a succint `1` or `0` output). The strong tendency of the BioMistral model to produce verbose outputs may hurt its viability for use as a hallucination classifier.

While the classification metrics above assess model reliability, system-level performance metrics provide insight into how feasible model deployment would be in a resource constrained environment. We now evaluate the effect of k-bit quantization on average latency, GPU load, peak memory usage, and inference throughput.

4.2.3 System Performance Metrics

Quantization, as a model compression technique, is generally applied to a model because of its effect on system-level metrics (e.g., reduce inference latency, peak GPU memory usage) [**?**]. Figures 4.6 through 4.9 show the impact of k-quantization on latency, GPU load, peak memory utilization, and throughput, respectively.

Figure 4.6 shows the **average inference latency** of various LLM families at different quantization levels.

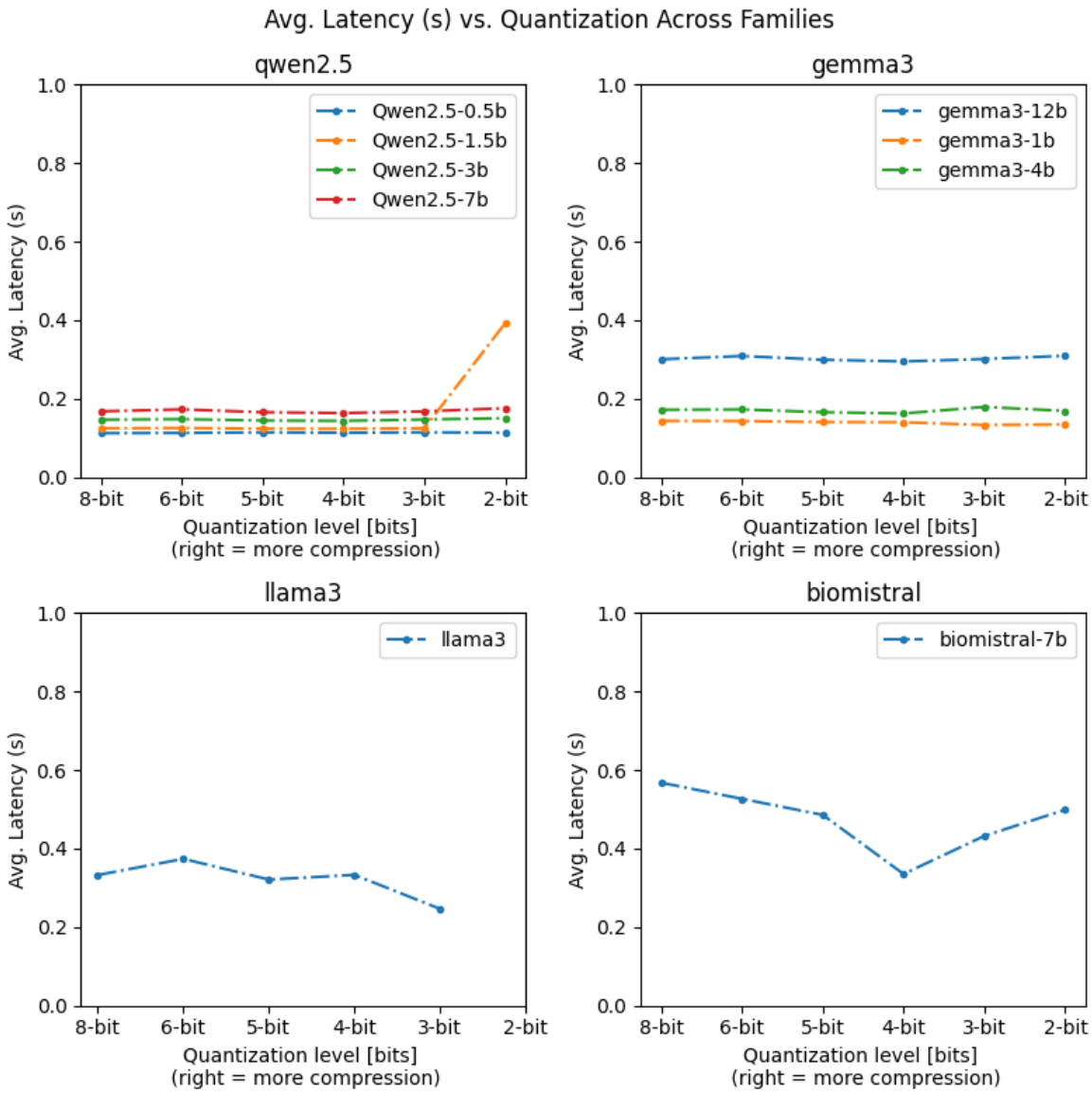


Figure 4.6: Average inference latency (seconds) as a function of k-quantization level. Lower K generally corresponds to faster inference (lower latency).

The general-reasoning models show stable latency across all quantization levels. An exception is Qwen2.5 1.5B model which shows increased latency at the 2-bit level. Larger models have systematically higher latency than smaller variants within the same family, though across families, the Qwen2.5 series has the lowest latency overall, followed by the Gemma3 family, Llama3, and BioMistral models.

Unlike the general-reasoning models, the medically fine-tuned Llama3 and BioMistral models,

show a decreasing average latency as quantization becomes more aggressive (K decreases).

Figure 4.7 shows the **average GPU load** (%) during inference across k-quantization levels. Lower GPU load is better for deployment, as it translates to lower energy usage and resource constraints. Understanding GPU load performance can help stakeholders assess how well a particular model-quantization configuration utilizes available compute resources. A low GPU load (e.g. with Gemma3-12B) may give room to increase throughput via processing prompts in parallel. On the other hand, a high GPU load (e.g. with BioMistral) may be a cause for concern, as it can lead to an increase in energy use and GPU temperature. This can induce performance throttling, a hardware safety mechanism that either reduces clockspeed [56] or shuts down to prevent damage [57].

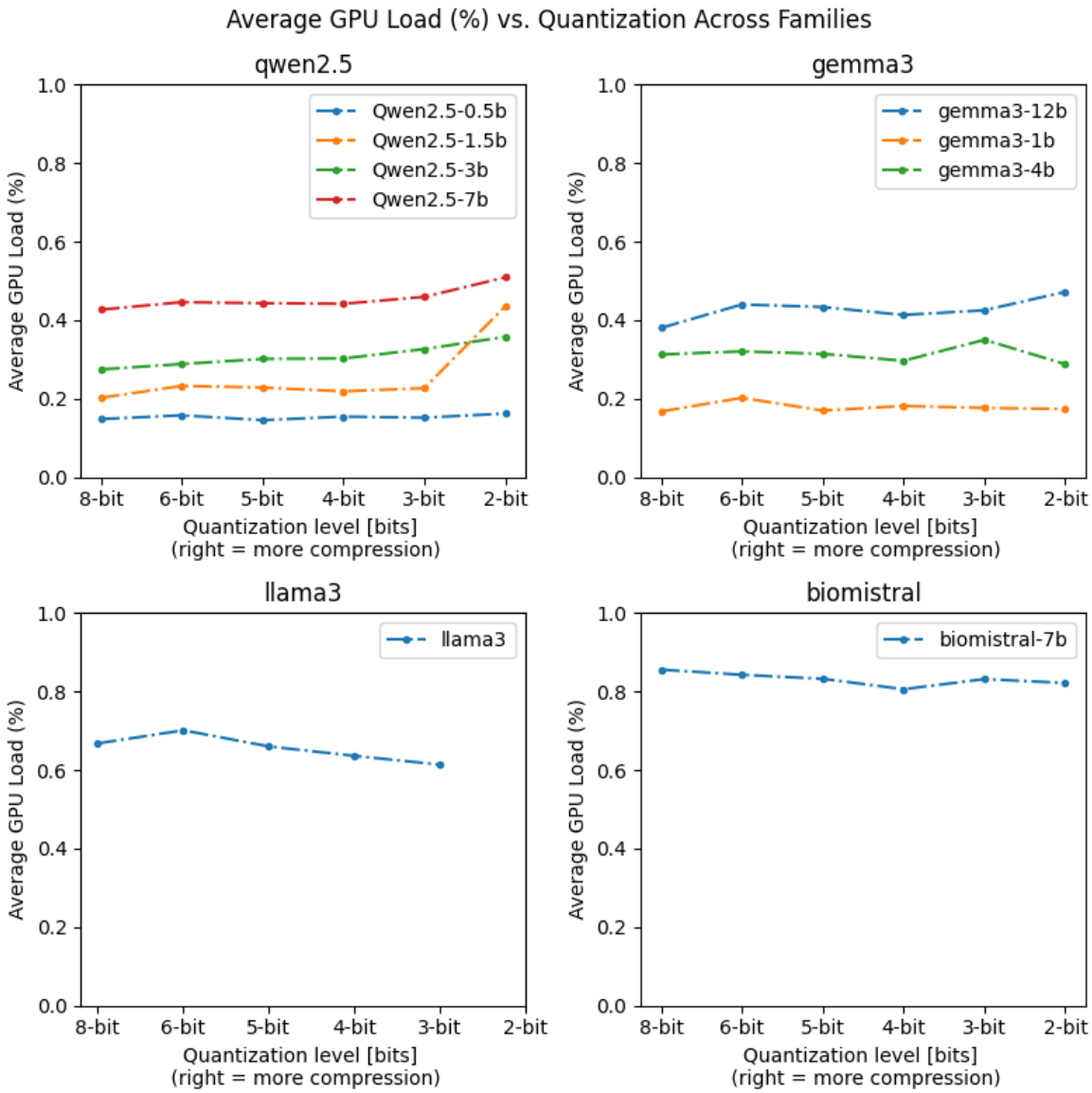


Figure 4.7: Average relative GPU load (%) versus K quantization level.

Medically fine-tuned models consistently demand a higher GPU load than their general-reasoning counterparts (e.g. the Gemma3 12B, 4B, and Qwen2.5 7B which are of similar size). (see Figure 4.6).

BioMistral 7B: GPU load remains relatively stable at around 80% for all quantization levels tested. This model demands a consistently higher GPU load than all other tested models.

Llama3 8B: Like BioMistral, the LLama3 model shows stable GPU demand (around 63%) for all quantization levels, with decreasing demand for quantization levels of 6-bits and lower.

Qwen2.5 Family:

- 0.5B: The GPU load is the least affected among all tested models as it changes the least across different quantization levels. It is also the smallest model, and demands the lowest GPU load.
- 1.5B: model also shows relatively stable GPU load, with a notable increase from around 20% at the 3-bit level to 40% at the 2-bit level.
- 3B and 7B: These models show a slow, increasing trend in GPU load as quantization becomes more aggressive.

Gemma3 Family:

- 1B: Stable GPU load, around 18%, with a positive spike to 21% at the 6-bit level.
- 4B: Stable GPU load around 30% from 8-bit to 4-bit, and the 2-bit level. However, there is a sudden increase in load at 3-bit the quantization level.
- 12B: Positive trend in GPU load as quantization becomes more aggressive. It increases from slightly under 40% at the 8-bit level to around 50% at the 2-bit level.

Summary of Insights into GPU Load: The trend for GPU load varies across model family and size. Medically fine-tuned models demand higher load overall, though BioMistral is stable whereas LLama3 shows a decreasing trend as quantization becomes more aggressive. For many Qwen2.5 and Gemma3 models, there is a slight positive trend in GPU load as quantization becomes more aggressive. This may indicate that the hardware used requires more computational overhead to process data passing through these two model families at lower bit-levels [36].
The **peak GPU memory** used during inference is presented in Figure 4.8. A lower precision representation of weights (e.g. 2-bit as opposed to 8-bits) requires fewer bits per LLM parameter. This implies that the memory usage should decrease for quantizations with lower bit representations [37, 36].

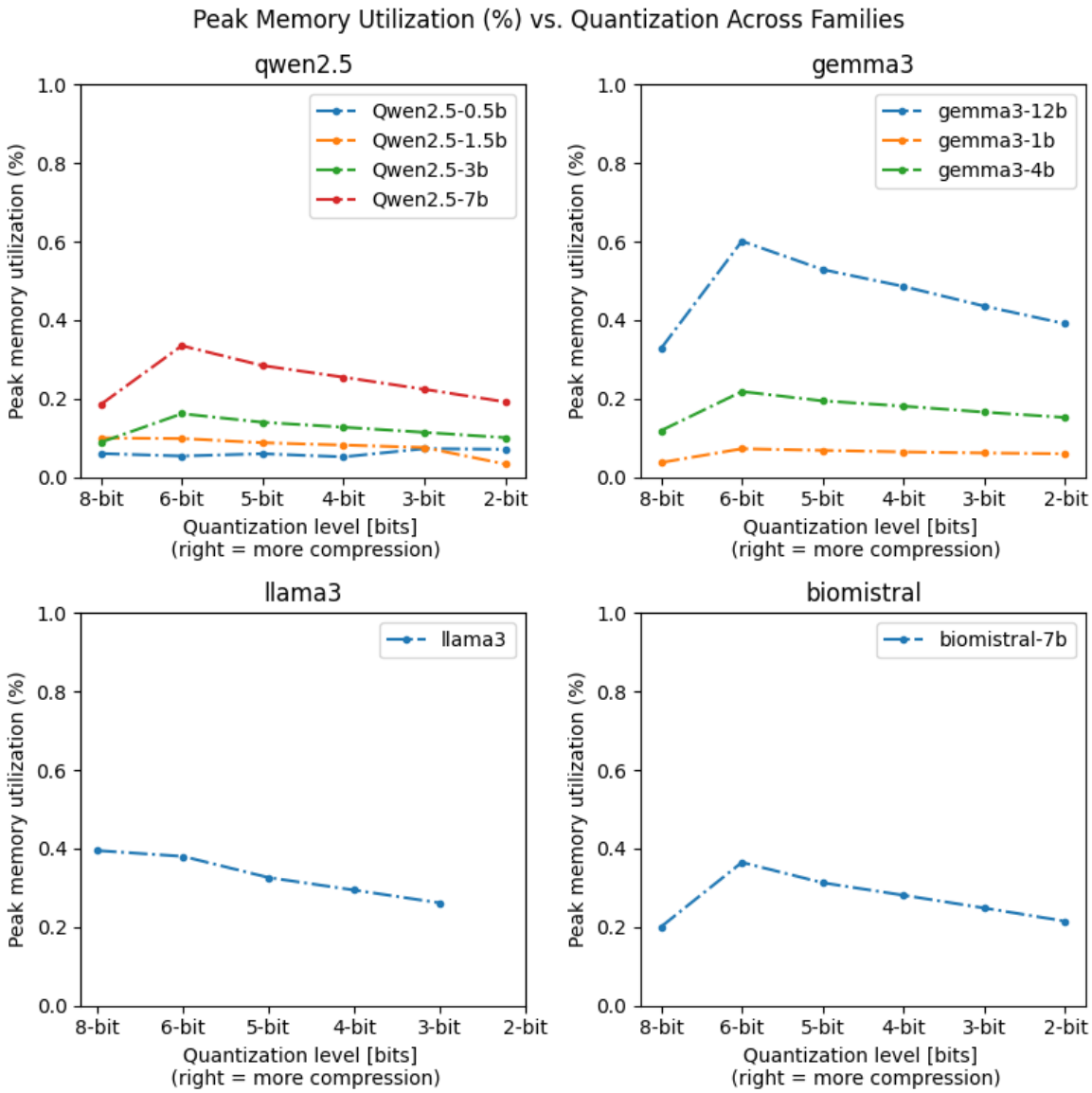


Figure 4.8: Peak GPU memory utilization (GB) vs. K-quantization level. Lower K generally reduces memory footprint.

Across all models, peak memory utilization generally increases as the number of quantization bits increases from 2 to 6 (with a spike at 6-bit quantization), and then interestingly, either plateaus (e.g., Llama3, Qwen2.5 1.5B and 0.5B models) or drops (e..g, Qwen2.5 7B, 3B; Gemma3 models, Biomistral). This behaviour indicates that intermediate bit-width representations of LLM weights may result in additional overhead needed by hardware to work with the bit-representation (e.g., for General Matrix Multiplication (GeMM)) [37, 36].

Finally, Figure 4.9 shows **inference throughput**, measured in prompts per second.

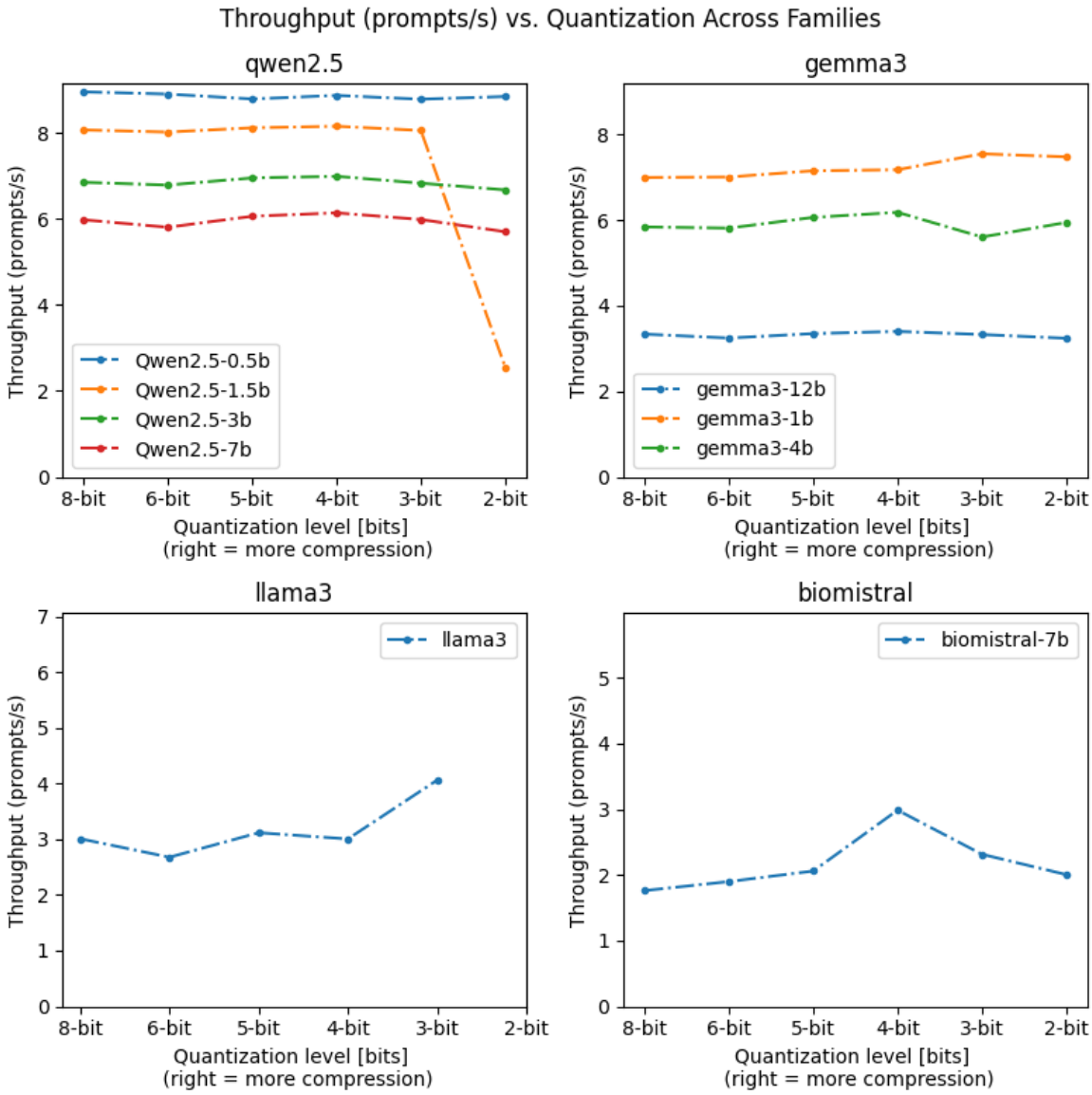


Figure 4.9: Inference throughput (prompts/sec) vs k-quantization level. Higher K generally leads to lower throughput due to increased latency.

The general-reasoning (Qwen2.5, Gemma3) models generally show stable throughput for all quantization levels. Larger models consistently classify fewer prompts per second than their smaller counterparts within the same family. This is consistent with the results of Figure 4.6, as throughput is inversely related to latency.

The medically fine-tuned models benefit more from quantization. Llama3 increases from 3 prompt-

s/second (8-bit quantization) to 4 prompts/second at the 2-bit quantization level, and BioMistral has more erratic performance, increasing in throughput from 8-bit (1.8 prompts/sec) to 4-bits (3 prompts/sec) and declining afterwards.

4.3 Performance Tradeoffs and Pareto Analysis

In the previous section, we showed that quantization affects multiple dimensions of model performance, and can do so in conflicting ways. For example, decreasing K improves (reduces) latency (Figure 4.6) and peak GPU memory usage (Figure 4.8), but does so typically at the cost of worsened (reduced) accuracy (Figure 4.1). Selecting the "best" model configuration (model family, k-bit quantization level) requires balancing competing objectives. In a more specific example, the best models with respect to latency (e.g. Qwen2.5 0.5B, 1.5B refer to Figure 4.6), are also be the worst performing models with respect to accuracy (e.g. Qwen2.5 0.5B, 1.5B refer to Figure 4.1). Pareto dominance, a concept from the field of multi-objective optimization, provides a method that can be used to analyze these tradeoffs [58][59][60][61].

4.3.1 Pareto Dominance and Optimality

If we only consider one criterion when choosing between models, we may arrive at a conclusion that contradicts the alternative conclusion we would have arrived at had we selected based on a second criteria. Both may be important metrics to a stakeholder interested in deploying an LLM-examiner, and a tradeoff plot can aid in visually determining the most suitable model(s) to use [60].

We draw upon the **Pareto dominance** relation to highlight optimal tradeoffs between competing objectives (e.g., accuracy and latency in Figure 4.3.2, accuracy and peak memory in Figure 4.3.3) [58].

Context: Formally, let $\mathbf{M}_1$ and $\mathbf{M}_2$ be two candidate models evaluated on K objective functions $o_1, ..., o_K$. Each objective function $o_k(\mathbf{M}_j)$ evaluates model $\mathbf{M}_j$ on a particular criterion (e.g. accuracy). We assume each o_k is either a positive objective (higher is better, e.g., accuracy) or a negative objective (lower is better, e.g., average latency).

Definition: We can say that model $\mathbf{M}_1$ **Pareto dominates** $\mathbf{M}_2$ if the following two conditions are held:

1. $\mathbf{M}_1$ is no worse than $\mathbf{M}_2$ on all objectives:

$$\forall k, \quad \begin{cases} o_k(\mathbf{M}_1) \geq o_k(\mathbf{M}_2), & \text{if } o_k \text{ is to be maximized} \\ o_k(\mathbf{M}_1) \leq o_k(\mathbf{M}_2), & \text{if } o_k \text{ is to be minimized} \end{cases}$$

2. $\mathbf{M}_1$ is strictly better in at least one objective:

$$\exists j \text{ such that } \begin{cases} o_j(\mathbf{M}_1) > o_j(\mathbf{M}_2), & \text{if } o_j \text{ is to be maximized} \\ o_j(\mathbf{M}_1) < o_j(\mathbf{M}_2), & \text{if } o_j \text{ is to be minimized} \end{cases}$$

To further illustrate our application of pareto dominance, we provide an example Python snippet in Appendix A.1. It identifies Pareto-optimal models based on the accuracy (positive) and peak memory usage (negative) objectives.
We identify all Pareto optimal points in our tradeoff plots to help identify which models offer the best compromise between competing objectives. We produce two tradeoff plots:

1. Figure 4.10 which compares accuracy and latency, and

2. Figure 4.11 which compares accuracy and memory.

4.3.2 Accuracy vs. Latency Tradeoff

Figure 4.10 shows the tradeoff between hallucination classification accuracy (a positive attribute; higher is better) and latency (a negative attribute; lower is better). Pareto-optimal models are highlighted with points that are circled and enlarged. These represent model-quantization configurations that achieve the best balance between accuracy and latency. In other words, no model perform strictly better than a Pareto-optimal model on both accuracy and latency.

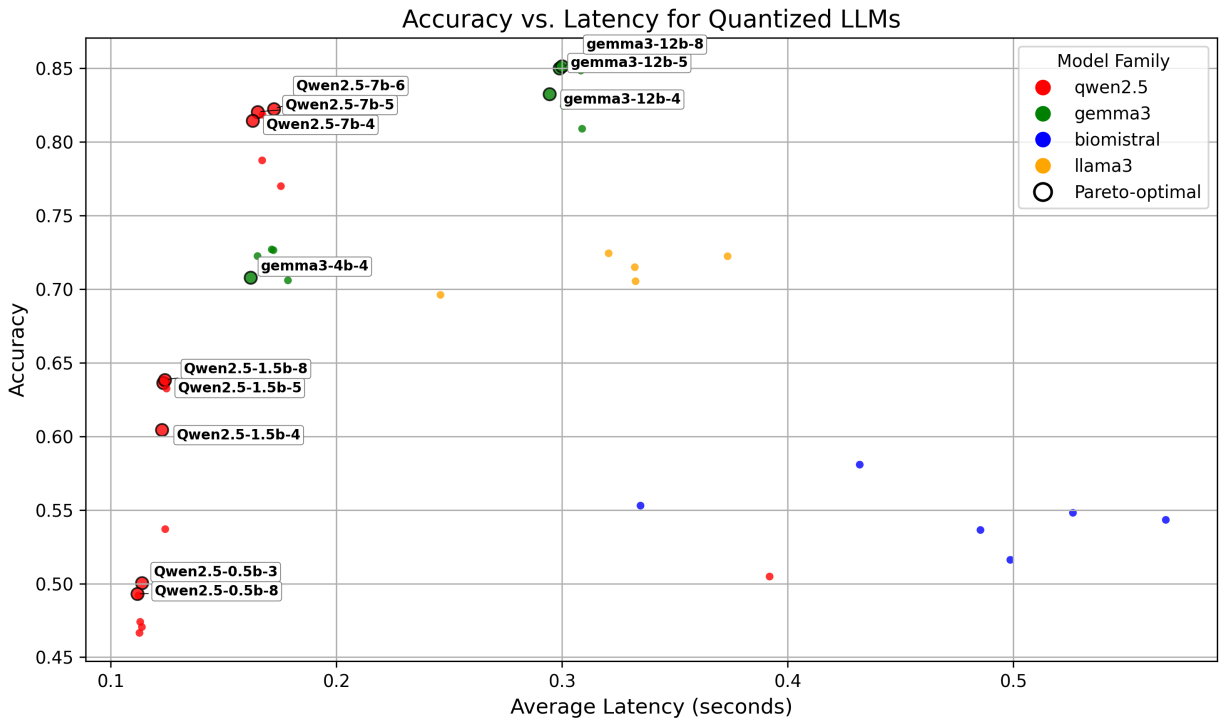


Figure 4.10: Accuracy vs. average latency tradeoff for quantized LLM-examiners. Circled points represent the pareto-optimal models, which offer the best latency for a given accuracy level (or vice versa).

Figure 4.10 above shows that the Pareto set comprises Qwen2.5 and Gemma3 models. The Llama and Biomistral models are low-performing in comparison. The Biomistral families shows that largest change in latency over different quantization schemes, but relatively low variance with respsect to accuracy. The family members of the Llama family are clustered closest together amongst varying quantization levels, and closest together compared to all other families. Quantization seems to have little effect on the Llama family relative to others. On the other hand, the Qwen2.5 series models is the most affected by quantization with respect to accuracy – the 0.5B, 8-bit quantized version sits on the pareto front with low latency (0.1s per prompt), but also accuracy below 50% (worse than random guessing), yet the 7B parameter, 4,5, and 6-bit quantizatiosn are clustered close together in the top right region of the plot, in the low latency, high accuracy region of the pareto frontier. Lastly, the Gemma3 models are clustered in three different regions in this tradeoff curve, corresponding to the small (1B) models in the lower-left region of the plot

(they are blocked by the annotations for the Qwen2.5-0.5B paremter models; refer to the Appendix), the 4B models in the upper left quadrant of the plot, and the 12B models in the top middle of the plot. These clusters indicate a different behaviour for the Gemma family – their accuracy latency-tradeoff seems to be primarily determined by model size, and not by extent of quantization. A limitation to this observation is that this behaviour may not extend to other quantization techniques like AWQ [22], GPTQ [23], LLM.int8() [21], and QServe [62], other compression methods (e.g., pruning, distillation), or combinations of compression methods [35]. Furthermore, these results may not extend to other tasks within the medical QA domain like Med-HALT [63], let alone other NLP tasks (e.g., abstractive summarization [**?**], financial reports). All in all, this tradeoff plot indicates that the different LLM families have distinct performance degradation behaviour with respect to accuracy and latency. In the context of medical QA on medical scientific literature (the task captured by MedHallu), a stakeholder trying to decide between these four models to use as an LLM-examiner, may want to choose between the Qwen2.5 and Gemma3 families, if latency is a major system-performance related constraint for them. The results from this study, are far from being able to cover the high-dimensional search space of various model families, compression schemes, tradeoff curves, NLP tasks, and data domains (e.g., medical, finance) that may be involved in deciding what LLM to go with for a real-world use case. Nevertheless, the results from this study motivate the act of comparing tradeoffs between model families – they exhibit distinct behaviours – a 7B Qwen2.5 model, is not the same as a 7B llama model, which is not the same as the Biomistral 7B models. Also, surprisingly, it is the general-reasoning language models which dominate the pareto frontier in the accuracy-latency tradeoff curve, and the medically-fine tuned models which lag behind in performance. This results agrees with the results in [4], who found that the general-reasoning models outperformed their medically fine-tuned counterparts.

4.3.3 Accuracy vs. Peak Memory

Figure 4.11 shows the tradeoff between accuracy (positive attribute) and peak GPU memory usage (negative attribute).

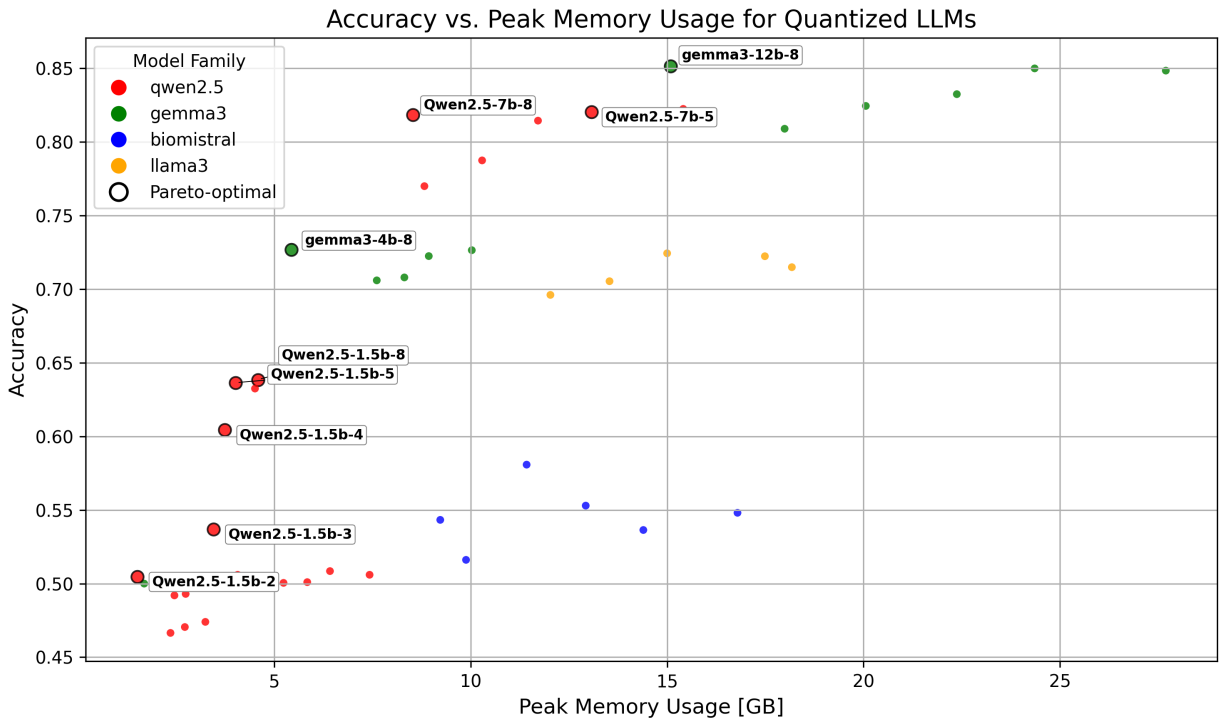


Figure 4.11: Shown above is the tradeoff curve between hallucination detection accuracy and peak memory usage [GB] of LLM-examiners. The pareto set comprises the Qwen2.5 and Gemma families. The low accuracy, low peak-memory usage region of the pareto frontier is primarily occupied by the Qwen series, whereas the high-accuracy medium peak memory usage region near the top-middle region of the plot has a mix of Qwen and Gemma3 models.

4.3.4 Sensitivity and Specificity: Breakdown over Hallucination Type

Chapter 5

Discussion

There are numerous ways in which prior works have studied the tradeoff between extent of LLM compression and model performance. For example, Zhu et. al show examples of comparing LLM compression rate and speedup [12], [64] compares compression ratio and perplexity, [35] compares LLM model size and accuracy.

Help stakeholders make informed trade-off decisions about which models to use in real-world constraints.

There are numerous modifications that a stakeholder interested in applying LLMs in settings where data privacy is important.

A limitation to keep in mind for stakeholders interested in applying LLMs in settings with sensitive data, is that compressing the model via quantization (i.e., to enable deployment on a device with relatively low memory compared to cloud or server compute alternatives) may negatively affect other modifications added to the LLM. For example, machine unlearning is a technique that can be applied to LLMs to bolster data privacy. However, a recent study by Zhang et al. [65] found that applying quantization can undo the effects of machine learning, if it was applied to an LLM before quantization.

Chapter 6

Conclusion

Appendix A

Appendix

A.1 Pareto Dominance Implementation

Listing A.1: Implementation of Pareto Dominance Check for Accuracy vs. Peak Memory

```
# Identify dominated models
for candidate in models:
    for challenger in models:
        if (
            # challenger not worse in all objectives
            challenger["accuracy"] >= candidate["accuracy"] and
            challenger["peak_mem"] <= candidate["peak_mem"] and
            (
                # and is better in at least 1 objective
                challenger["accuracy"] > candidate["accuracy"] or
                challenger["peak_mem"] < candidate["peak_mem"]
            )
        ):
            candidate["dominated"] = True
```

A.2 Pareto Plot

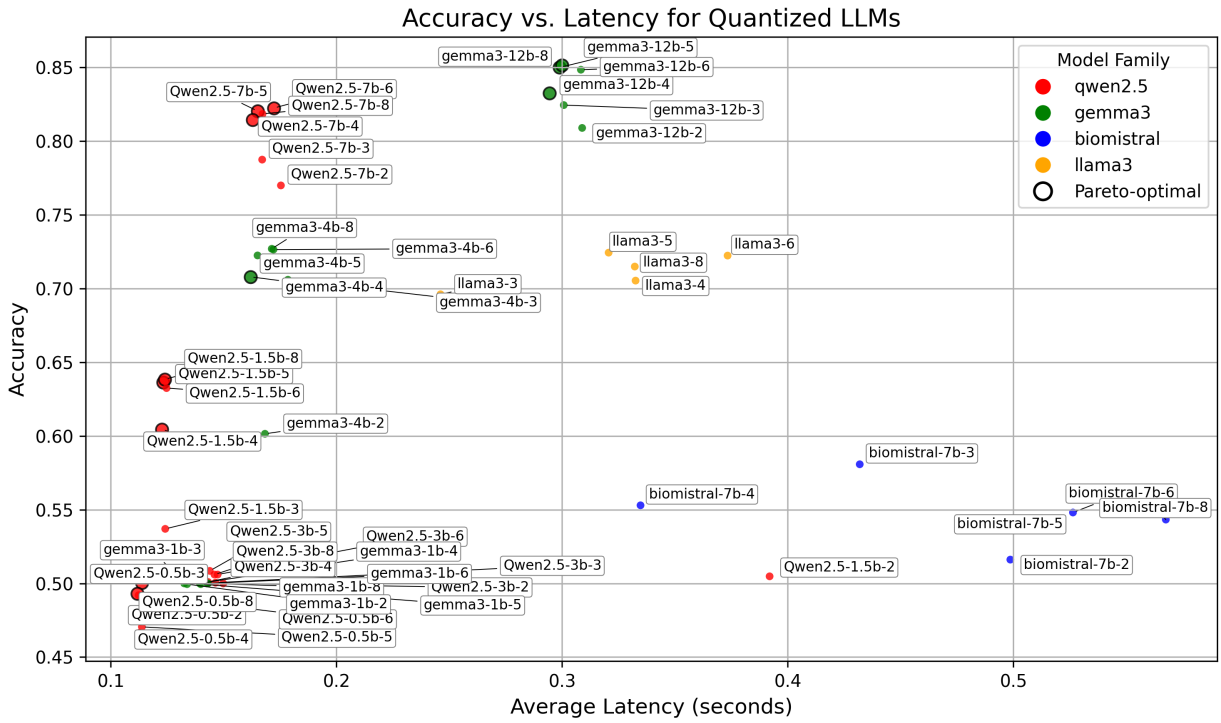


Figure A.1: Accuracy vs. average latency tradeoff for quantized LLM-examiners. Circled points represent the pareto-optimal models, which offer the best latency for a given accuracy level (or vice versa). This plot shows annotations for all models, not just the pareto dominant ones.

A.3 Runtime Logs and Dequantization Warnings

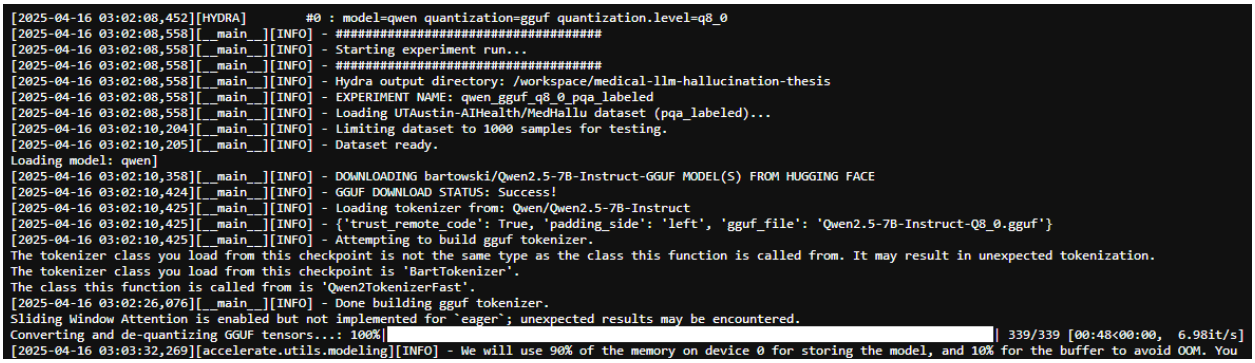


Figure A.2: Terminal log showing Hugging Face runtime de-quantizing GGUF tensors before inference, reducing potential memory benefits.

A.4 Vast.AI GPU Instance Details

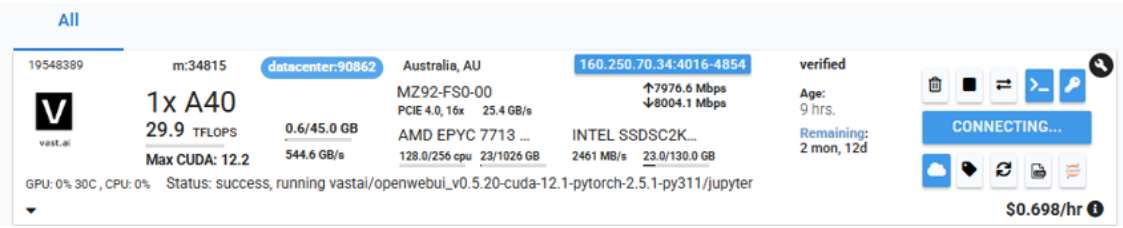


Figure A.3: Screenshot of Vast.AI dashboard showing rented A40 GPU instance at a rate of CAD $0.698/hr used for all final experiments.

A.5 Cost Estimation Script

Total Inference Time Estimation Script:

```
import os
import json

total_seconds = 0
root_dir = 'results/'
```

```
for subdir, _, files in os.walk(root_dir):
    for file in files:
        if file.endswith(".json"):
            path = os.path.join(subdir, file)
            with open(path) as f:
                data = json.load(f)
                try:
                    total_seconds += data['metrics']['latency_report']%['total_exp_infer
                except KeyError:
                    pass  # some folders might be incomplete or failed runs
print(f"Total inference time across all experiments: {round(total_seconds/3600, 2)} hour
```

This script helps compute the total runtime across all experiments, enabling estimation of the total experiment cost via:

$$\text{Total Cost (CAD)} = \text{Total Time (hours)} \times 0.698$$

A.6 Prompt Parsing Logic

Minimal Parsing Function (Python):

```
def extract_prediction(text: str) -> str:
    for token in text.strip()[:10]:
        if token in ['0', '1']:
            return token
    return None
```

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